



PaperHawk

Multi-agent document intelligence on AMD Instinct MI300X.
Built by engineers who ship.

RAG retrieves. Audit finds.

Today's RAG chatbots can do the first. They cannot do the second.

What RAG does well

Chunk a document. Embed the chunks. Retrieve top-K passages.
Generate an answer with the retrieved context.

Great for FAQ chatbots. Great for Q&A on a single document.

What auditors actually need

"Does the supplier in Invoice #7 match the vendor in PO #3? Is the VAT rate consistent across the package? Any change-of-control clauses? Sanctions hits?"

These questions live in the *relationship* between documents — not in any single chunk.

WHAT WE BUILT

A multi-agent system. Not a retrieval pipeline.

LangGraph 0.6-native. Production-shaped. Open source under MIT.

4

Compiled
graphs

6

Reusable
subgraphs

14

Deterministic
domain checks

5+1

Anti-halluc
layers

5

Agentic
chat tools

Send-API parallelism · `AsyncSqliteSaver` checkpointer · `configurable_alternatives` provider stack (vLLM / Ollama / dummy) · multi-agent DD assistant with 4 specialists
+ supervisor + synthesizer · Streamlit 5-tab UI · 61 tests passing in CI without an LLM.

Five steps. **End-to-end.**

Every step is a typed Pydantic-state node. Every LLM call has structured output.

1

Ingest

PDF · DOCX · image. Vision-first
OCR fallback for scanned
pages.

2

Classify

6-way doc-type classifier. ISA
500 evidence-quality score.

3

Extract

Pydantic schema per doc-type.
_quotes + _confidence per field.

4

Cross-ref

3-way matching. Package-
level analyzer. DD multi-agent.

5

Risk + Report

14 checks (parallel Send) · LLM
ensemble · 3-layer filter · DOCX
export.

*On AMD MI300X with Qwen 2.5 14B: **30–90 seconds** end-to-end per package.*

Fourteen rules. **In Python.**

Every check is a typed Protocol, not a prompt. Run in parallel via the LangGraph Send API.

TIER A — AUDIT · 6 CHECKS

ISA 500 Evidence hierarchy

ISA 320 Materiality threshold

ISA 240 Duplicate invoice detector

ISA 240 Rounded-amount anomaly

Tax-ID CDV mod-11 checksum

Mandatory fields Invoice completeness

TIER B — COMPLIANCE · 4 CHECKS

GDPR Art. 28 Sub-processor clause

AML / Sanctions EU + OFAC fuzzy match

M&A red flag Change-of-control · auto-renewal

Disproportionality Penalty-vs-value ratio

TIER C — STANDARDS · 4 CHECKS

Incoterms 2020 11-rule recognizer

IFRS / GAAP Goodwill + lease anomaly

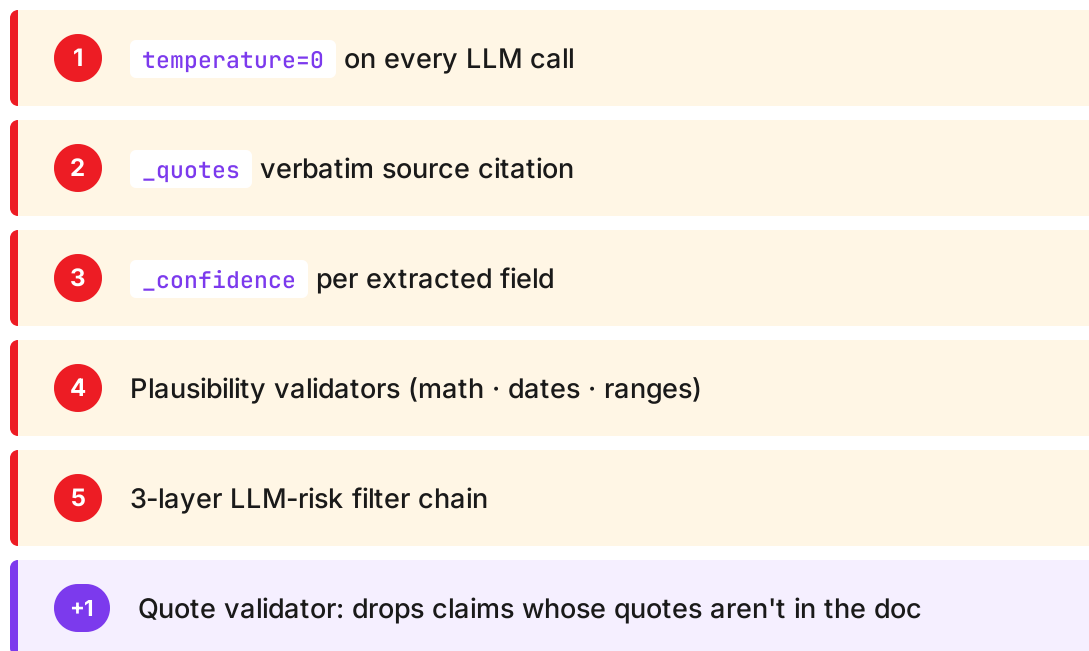
Math validation Net + VAT + gross

Contract completeness 6-key-clause check

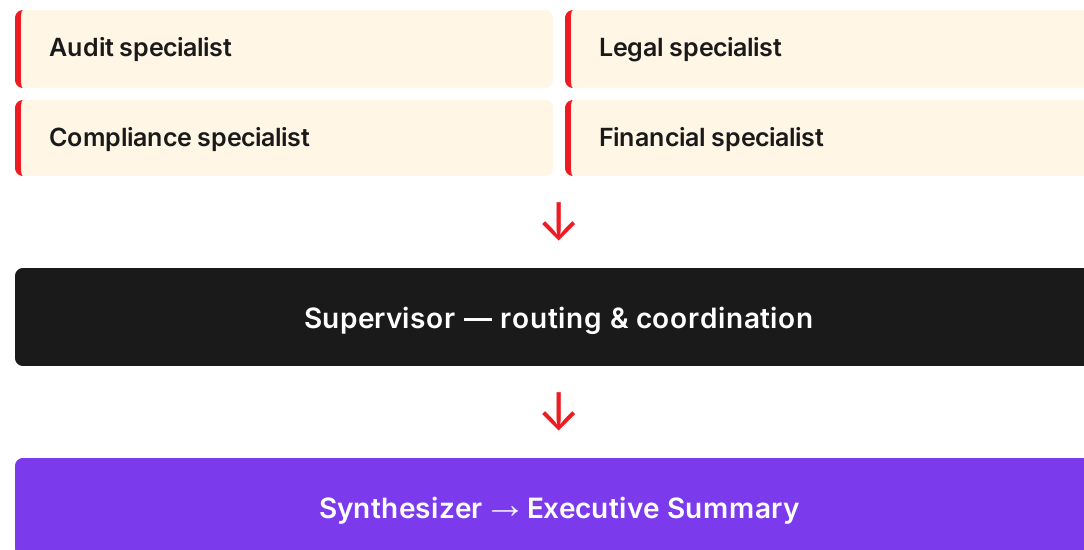
Jurisdiction-aware: locale-specific rules trigger only on locale-tagged inputs. Universal rules run everywhere.

Anti-halluc **5+1**. DD **multi-agent**.

5+1 LAYERS, EVERY OUTPUT



DD SUPERVISOR PATTERN



Four specialists read the same package independently. The supervisor coordinates routing. The synthesizer writes a 3-paragraph executive brief with cited red flags.

Qwen on **AMD MI300X** via vLLM.

192 GB HBM3. ROCm-native. Open-source models, end-to-end.

Streamlit · 5-tab UI

Upload · Results · Chat · DD · Report

LangGraph 0.6 orchestration

4 graphs · 6 subgraphs · Send API · AsyncSqliteSaver

Qwen 2.5 14B Instruct (open source)

tool-calling · structured-output · multilingual

vLLM continuous batching

--api-key · --max-model-len 32768 · OpenAI-compatible

AMD Instinct MI300X · ROCm

192 GB HBM3 · BF16 / FP8 · AMD Developer Cloud

Hugging Face Spaces deploy

lablab-ai-amd-developer-hackathon · Streamlit SDK

SEE IT IN ACTION

Three **one-click** demos.

Bundled in the repo. Drivable from the Streamlit Upload tab in 30 seconds.

Audit Demo

Three invoices from the same supplier. The March one is 50% pricier than January and February.

→ ISA 240 over-billing pattern flagged with cited line items.

DD Demo

NDA + service agreement + amendment in an acquisition scenario.

→ Hidden change-of-control + auto-renewal red flags.

Compliance Demo

Two contracts; one is missing GDPR Article 28 sub-processor language.

→ Domain check #8 detects the gap with regulatory citation.

On AMD MI300X with Qwen 2.5 14B Instruct: **30–90 seconds** per package · end-to-end · with citations.

Built for builders.

MIT licensed. Reproducible from a clean clone. No closed weights, no proprietary extensions.

/ 01

Open source · MIT

Public GitHub repo. No "training data not included" footnotes. Clone it, run it, fork it. The whole codebase is yours to read.

/ 02

Reproducible

Same stack from laptop to MI300X.

`infra/vllm/Dockerfile` + `serve.sh` + `requirements.txt`. One command, one container.

/ 03

Battle-tested

61 tests passing in CI without any LLM. Deterministic dummy provider for CI; vLLM and Ollama for everything else.

github.com/nandorfivince/paperhawk

| **HF Space:** [lablab-ai-amd-developer-hackathon/paperhawk](https://huggingface.co/lablab-ai-amd-developer-hackathon/paperhawk)

| **License:** MIT

THE TEAM

Three engineers. One **shipped product**.

We've shipped together for nearly a decade. PaperHawk is what happens when domain knowledge, engineering rigor, and product instinct meet on the same codebase.

LEAD · LANGGRAPH · AMD ADAPTATION

Vince Nándorfi

System architecture, domain research, ROCm/vLLM adaptation, testing. PaperHawk's blueprint and the AMD-edition rewrite.

ENGINEERING · DEVOPS

Tamás Vitai

Senior++ engineer. Implementation, infrastructure, integration testing. Where the code meets the runtime.

ENGINEERING · ALGORITHMS

Gábor Murcsik

Engineering rigor. Algorithmic precision. Senior systems thinking, sharpened over years of complex production builds.

Beyond simple **RAG**. Built to **ship**.